Practical 10

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

```
GNU nano 8.7                                   sstf.c                                   Modified
#include <stdio.h>
#include <stdlib.h>

int main() {
    int req[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;

    int visited[8] = {0};
    int head = 53;
    int total_movement = 0;

    printf("Order of execution:\n");

    for(int i = 0; i < n; i++) {
        int min = 10000;
        int index = -1;

        for(int j = 0; j < n; j++) {
            if(!visited[j]) {
                int distance = abs(head - req[j]);

                if(distance < min) {
                    min = distance;
                    index = j;
                }
            }
        }

        visited[index] = 1;
        total_movement += min;
        head = req[index];

        printf("%d -> ", head);
    }

    printf("\n\nTotal Head Movement = %d\n", total_movement);

    return 0;
}

^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo      M-A Set Mark   M-] To Bracket
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^B Where Was
```

```
FOIN@LAPTOP-MC73RDEH MSYS ~
$ nano sstf.c

FOIN@LAPTOP-MC73RDEH MSYS ~
$ gcc sstf.c -o sstf

FOIN@LAPTOP-MC73RDEH MSYS ~
$ ./sstf
Order of execution:
41 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 14 ->

Total Head Movement = 323

FOIN@LAPTOP-MC73RDEH MSYS ~
$ |
```